```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

pd.set_option('display.max_columns', None)

print("Libraries loaded successfully")
```

```
Libraries loaded successfully
```

```
from google.colab import files

uploaded = files.upload()
```

Choose Files ecommerce..._ratios.csv
ecommerce_customer_data_custom_ratios.csv(text/csv) - 21418688 bytes, last modified: 25/5/2026 - 100% done

```
Saving ecommerce_customer_data_custom_ratios.csv to ecommerce_customer_data_c
```

```
df = pd.read_csv("ecommerce_customer_data_custom_ratios.csv")

df.head()
```

	Customer ID	Purchase Date	Product Category	Product Price	Quantity	Total Purchase Amount	Payment Method	Custor
0	46251	2020-09-08 09:38:32	Electronics	12	3	740	Credit Card	
1	46251	2022-03-05 12:56:35	Home	468	4	2739	PayPal	
2	46251	2022-05-23 18:18:01	Home	288	2	3196	PayPal	
3	46251	2020-11-12 13:13:29	Clothing	196	1	3509	PayPal	
4	13593	2020-11-27 17:55:11	Home	449	1	3452	Credit Card	

```
# Dataset information

print("Shape of dataset:", df.shape)
```

```
print("\nColumn data types:")
print(df.dtypes)

print("\nMissing values:")
print(df.isnull().sum())
```

```
Shape of dataset: (250000, 13)

Column data types:
Customer ID                int64
Purchase Date             object
Product Category          object
Product Price              int64
Quantity                   int64
Total Purchase Amount      int64
Payment Method            object
Customer Age               int64
Returns                  float64
Customer Name             object
Age                        int64
Gender                    object
Churn                      int64
dtype: object

Missing values:
Customer ID                  0
Purchase Date                0
Product Category             0
Product Price                0
Quantity                     0
Total Purchase Amount        0
Payment Method               0
Customer Age                 0
Returns                  47596
Customer Name                0
Age                          0
Gender                       0
Churn                        0
dtype: int64
```

```
# Summary statistics

print("Mean values:")
print(df.mean(numeric_only=True))

print("\nMedian values:")
print(df.median(numeric_only=True))

print("\nMode values:")
print(df.mode().iloc[0])
```

```
Mean values:
Customer ID              25004.036240
Product Price              254.659512
Quantity                     2.998896
Total Purchase Amount     2725.370732
Customer Age                43.940528
```

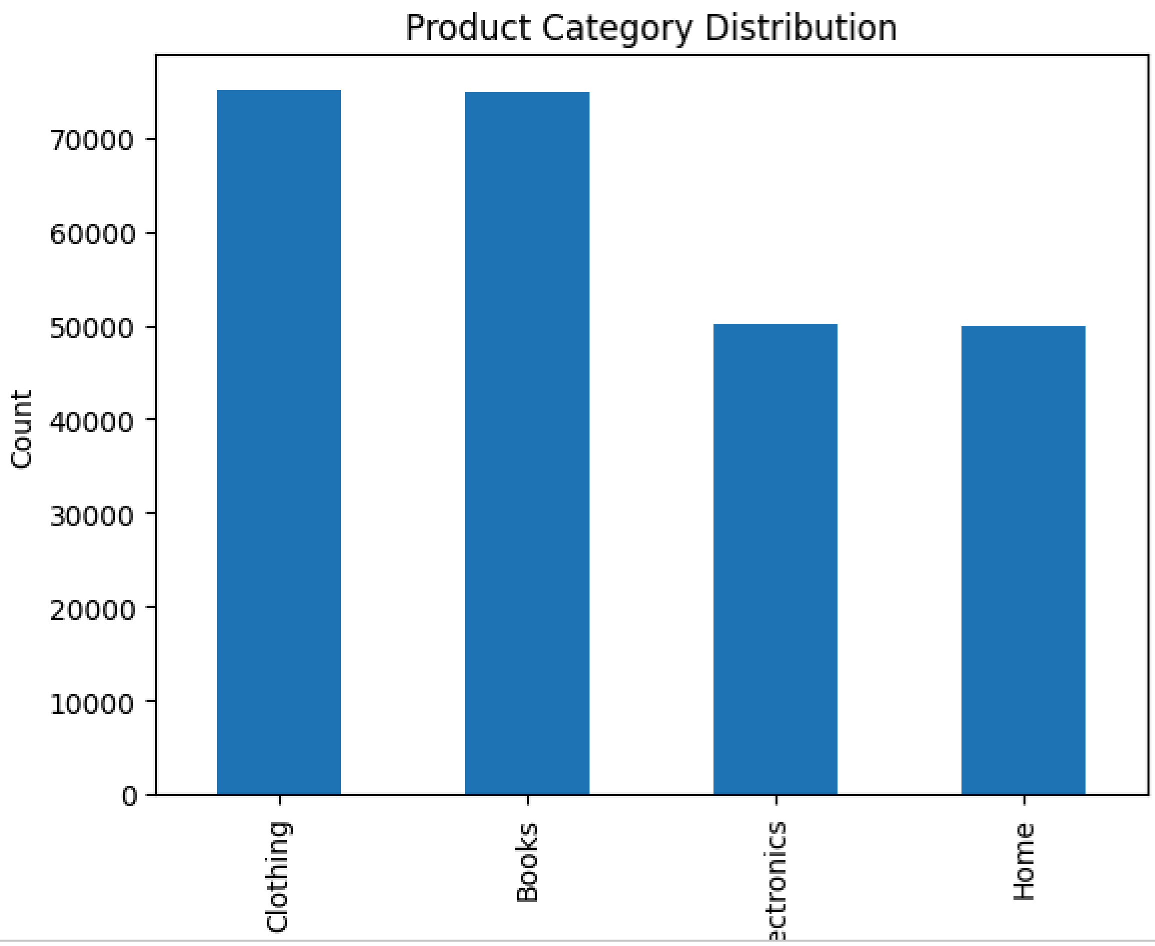
```
Returns                    0.497861
Age                       43.940528
Churn                      0.199496
dtype: float64

Median values:
Customer ID            25018.0
Product Price            255.0
Quantity                   3.0
Total Purchase Amount   2724.0
Customer Age              44.0
Returns                    0.0
Age                       44.0
Churn                      0.0
dtype: float64

Mode values:
Customer ID                             36437
Purchase Date             2022-05-08 12:58:55
Product Category                     Clothing
Product Price                           100.0
Quantity                                  1.0
Total Purchase Amount                  2786.0
Payment Method                    Credit Card
Customer Age                             58.0
Returns                                   0.0
Customer Name                   Michael Smith
Age                                      58.0
Gender                                 Female
Churn                                     0.0
Name: 0, dtype: object
```

```
# Data Cleaning

# Check duplicate rows
print("Duplicate rows:", df.duplicated().sum())

# Fill missing values in Returns with 0
df["Returns"] = df["Returns"].fillna(0)

# Remove duplicate age column
df = df.drop("Customer Age", axis=1)

print("\nMissing values after cleaning:")
print(df.isnull().sum())
```

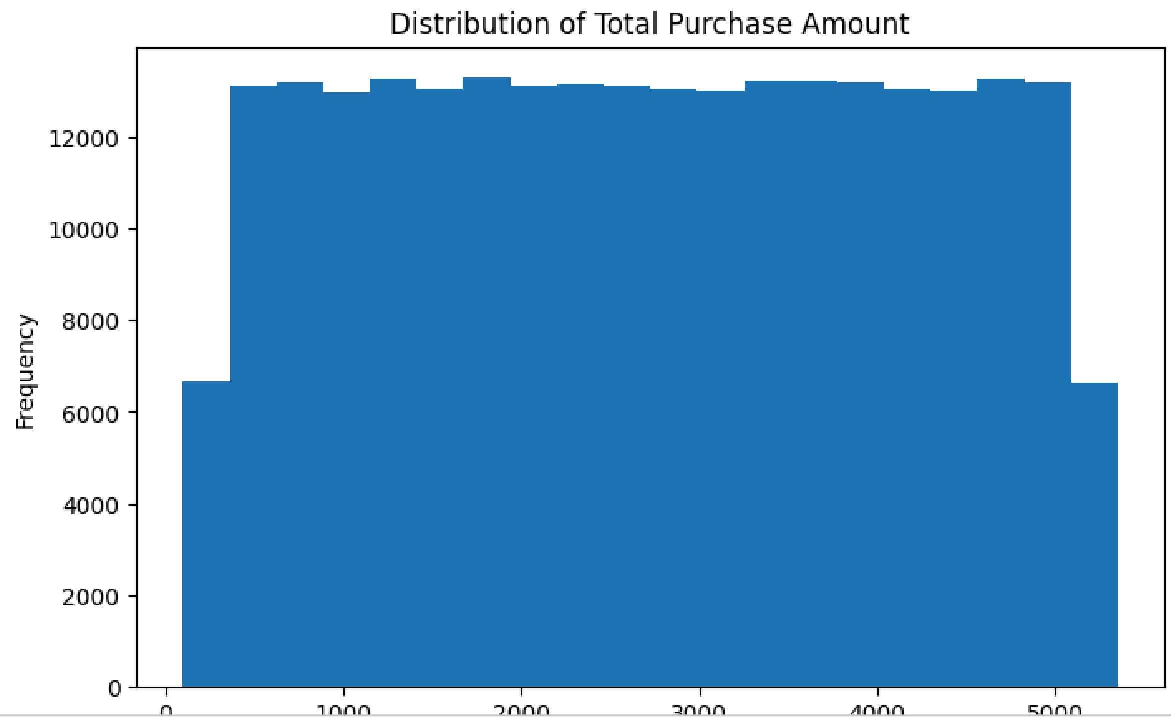
```
Duplicate rows: 0

Missing values after cleaning:
Customer ID              0
Purchase Date            0
Product Category         0
Product Price            0
Quantity                 0
Total Purchase Amount    0
Payment Method           0
Returns                  0
Customer Name            0
```

```
Age                      0
Gender                   0
Churn                    0
dtype: int64
```

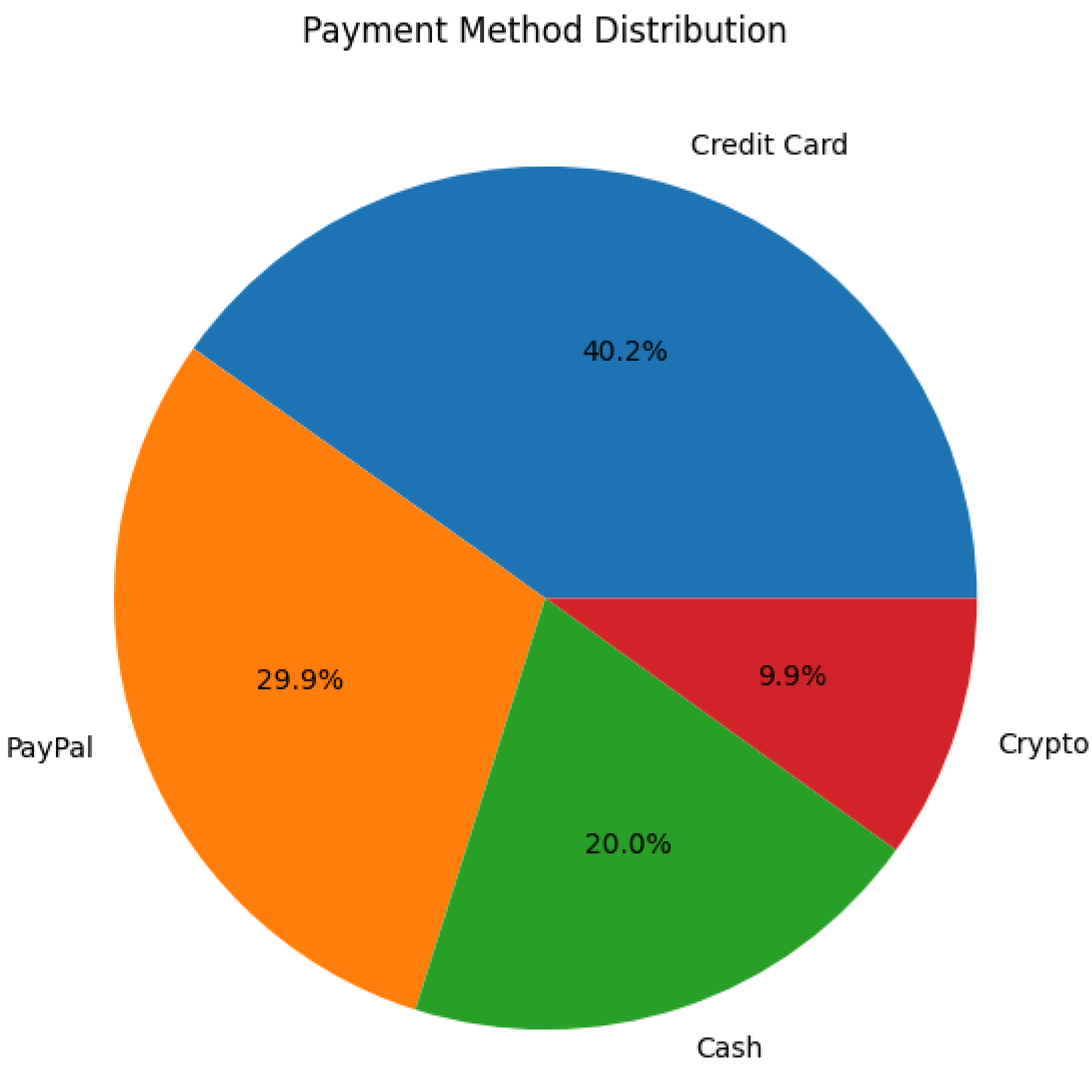
```
# Product Category Distribution (Bar Chart)

df["Product Category"].value_counts().plot(kind="bar")

plt.title("Product Category Distribution")
plt.xlabel("Product Category")
plt.ylabel("Count")

plt.show()
```




```
# Purchase Amount Distribution (Histogram)

plt.figure(figsize=(8,5))

plt.hist(df["Total Purchase Amount"], bins=20)

plt.title("Distribution of Total Purchase Amount")
plt.xlabel("Total Purchase Amount")
plt.ylabel("Frequency")

plt.show()
```




```
# Payment Method Distribution (Pie Chart)

payment_counts = df["Payment Method"].value_counts()

plt.figure(figsize=(7,7))

plt.pie(payment_counts,
        labels=payment_counts.index,
        autopct='%1.1f%%')

plt.title("Payment Method Distribution")

plt.show()
```




```
# Convert purchase date into datetime format

df["Purchase Date"] = pd.to_datetime(df["Purchase Date"])

# Create reference date
reference_date = df["Purchase Date"].max()

# RFM table
rfm = df.groupby("Customer ID").agg({
    "Purchase Date": lambda x: (reference_date - x.max()).days,
    "Customer ID":"count",
    "Total Purchase Amount":"sum"
})

rfm.columns = ["Recency","Frequency","Monetary"]

rfm.head()
```

Customer ID	Recency	Frequency	Monetary
1	57	1	3491
2	298	3	7988
3	88	8	22587
4	126	4	8715
5	170	8	12524

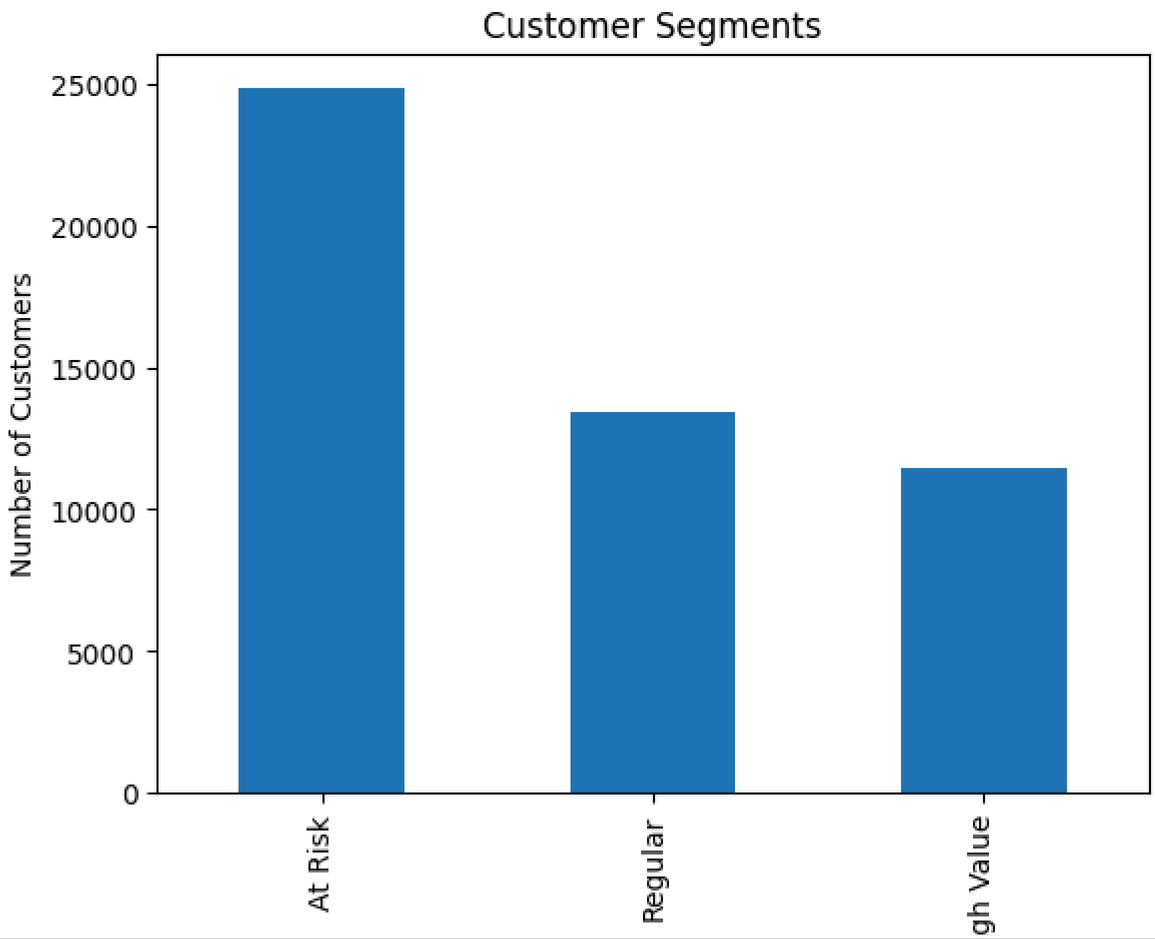
Next steps: Generate code with rfm New interactive sheet

```
# Customer segmentation

rfm["Segment"] = "Regular"

rfm.loc[
    (rfm["Frequency"] > rfm["Frequency"].median()) &
    (rfm["Monetary"] > rfm["Monetary"].median()),
    "Segment"
] = "High Value"

rfm.loc[
    (rfm["Recency"] > rfm["Recency"].median()),
    "Segment"
] = "At Risk"

rfm["Segment"].value_counts()
```

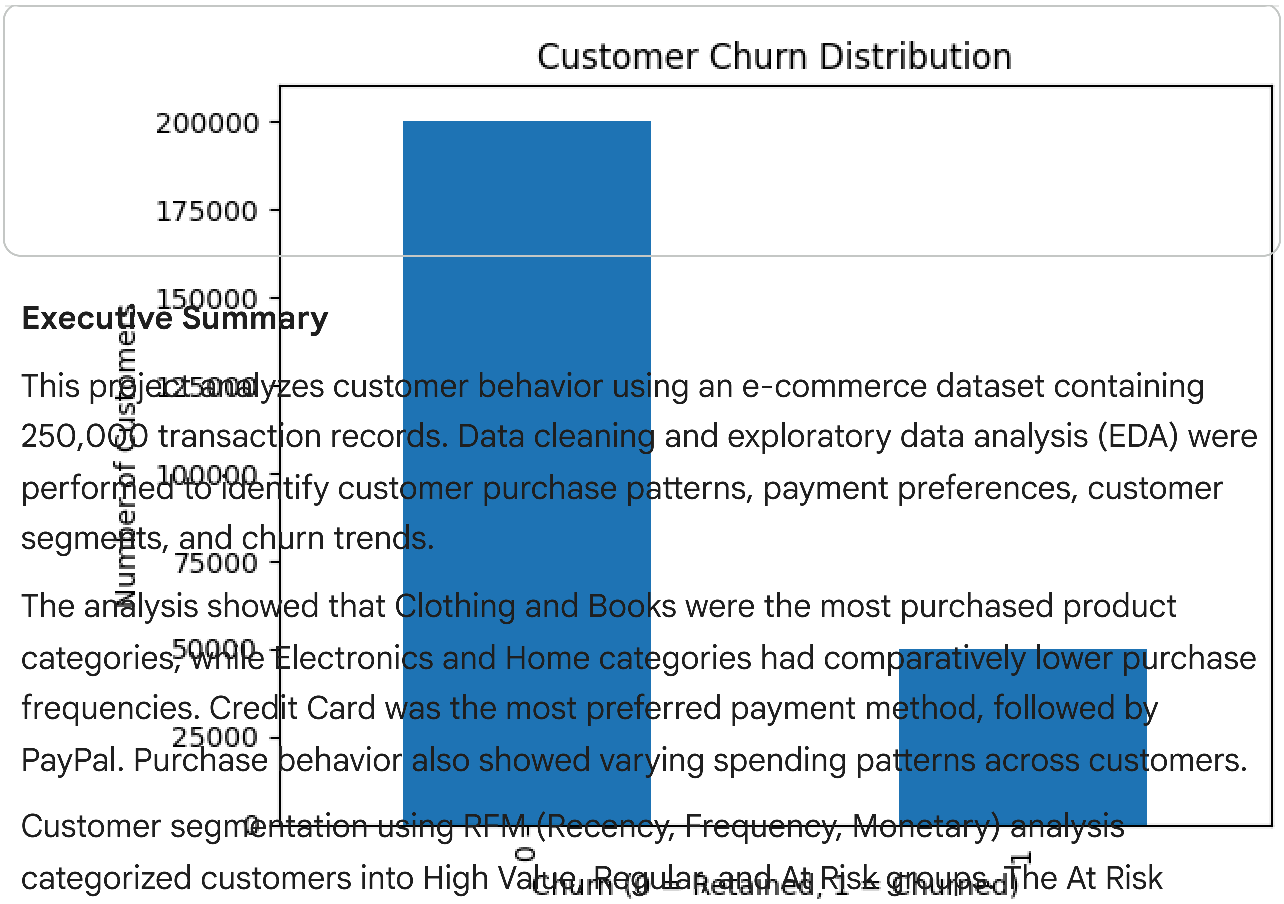
Segment	count
At Risk	24815
Regular	13392
High Value	11466

dtype: int64

```
# Customer Segment Visualization

rfm["Segment"].value_counts().plot(kind="bar")

plt.title("Customer Segments")
plt.xlabel("Segment")
plt.ylabel("Number of Customers")

plt.show()
```




```
# Churn Distribution

df["Churn"].value_counts().plot(kind="bar")

plt.title("Customer Churn Distribution")
plt.xlabel("Churn (0 = Retained, 1 = Churned)")
plt.ylabel("Number of Customers")

plt.show()
```



Executive Summary

This project analyzes customer behavior using an e-commerce dataset containing 250,000 transaction records. Data cleaning and exploratory data analysis (EDA) were performed to identify customer purchase patterns, payment preferences, customer segments, and churn trends.

The analysis showed that Clothing and Books were the most purchased product categories, while Electronics and Home categories had comparatively lower purchase frequencies. Credit Card was the most preferred payment method, followed by PayPal. Purchase behavior also showed varying spending patterns across customers.

Customer segmentation using RFM (Recency, Frequency, Monetary) analysis categorized customers into High Value, Regular, and At Risk groups. The At Risk segment represented the largest customer group, indicating a need for stronger retention strategies. Churn analysis also showed that although customer retention was high, a considerable number of customers still left the platform.

The findings suggest that Alfido Tech can improve customer engagement through targeted retention campaigns, loyalty programs, and category-specific promotions.

Actionable Recommendations for Alfido Tech

1. Focus retention campaigns on At Risk customers using personalized offers and reminder campaigns.
2. Implement loyalty programs for High Value customers to increase repeat purchases and customer lifetime value.
3. Increase promotions for Electronics and Home products to improve category